AQ2.2: Activity Questions 2- Not Graded

This assignment will not be graded and is only for practice.
Level 1:
1 point
Consider a system of linear equations as follows:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A (*i.e.*, $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b, for $i = 1, 2, 3$. If $det(A_{x_i}) = 0$ for $i = 1, 2, 3$, then which of the following is (are) true?

- [ ] $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ must be a solution.
- [ ] $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a solution if $det(A) \neq 0$.
- [ ] If $det(A) = 0$, then we cannot conclude about the solution using Cramer's rule.
- [ ] If $det(A) \neq 0$, then $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$x = \begin{bmatrix}0\\0\\0\end{bmatrix}$ is a solution if $det(A) \neq 0$.

If $det(A) = 0$, then we cannot conclude about the solution using Cramer's rule.

If $det(A) \neq 0$, then $b = \begin{bmatrix}0\\0\\0\end{bmatrix}$.

Consider a system of linear equations

$$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$

Let matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix}1 & 0 & 1\\-1 & 1 & -1\\0 & -1 & 1\end{bmatrix}$, $x = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$, and $b = \begin{bmatrix}1\\1\\1\end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A (i.e., $\begin{bmatrix}a_{1i}\\a_{2i}\\a_{3i}\end{bmatrix}$) by b, for $i = 1, 2, 3$.

(Use the above information to answer questions 2 and 3)

1 point

Choose the set of correct options

☐ $A_{x_1} = \begin{bmatrix}1 & 1 & 0\\-1 & 1 & -1\\0 & -1 & 1\end{bmatrix}$

☐ $A_{x_1} = \begin{bmatrix}1 & 0 & 1\\1 & 1 & -1\\1 & -1 & 1\end{bmatrix}$

☐ $A_{x_2} = \begin{bmatrix}1 & 0 & 1\\-1 & 1 & -1\\0 & -1 & 1\end{bmatrix}$

☐

$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

1 point
Choose the set of correct options.

- [ ] $x_1 = -2$.
- [ ] $x_2 = -2$.
- [ ] $x_3 = 3$.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x_1 = -2$.
$x_3 = 3$.

Level 2:

Consider the system of linear equations $Ax = b$, where $A = \begin{bmatrix} 1 & a & 0 \\ a & 1 & 2 \\ 1 & 0 & a \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, and the solution for x is partially known. What is the value of a^2, if $a \neq 0, \sqrt{3}, -\sqrt{3}$

is given? [Hint: Observe that the second row of the vector x is given as 0, which implies that x_2 is known, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider the system of linear equations $Ax = b$, where $A = \begin{bmatrix} 1 & 2a+3 \\ 3a+2 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and the solution for x is partially known.

What is the value of $3a$, if $det(A) \neq 0$ is given?

No, the answer is incorrect.
Score: 0